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RESEARCH ARTICLE

AI-Driven Smart Shopping Carts With Real-Time Tracking and Inventory Forecasting for Enhanced Retail Efficiency

MUHAMMAD IMRAN ZULFIQAR¹, AYESHA KHALID², ABUBAKR SIDDIG³,
MUHAMMAD JUNAID NAWAZ⁴, AND SALIM SAAY⁵

¹School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China

²Department of Computer Science and Information Technology, University of Jhang, Jhang 35200, Pakistan

³School of Computer Science, Technological University Dublin, Dublin 7, D07 EWV4 Ireland

⁴Department of Physical and Numerical Sciences, Qurtuba University of Science and Information Technology, Dera Ismail Khan 29111, Pakistan

⁵Department of Computer Science and Information Systems, University of Limerick, Limerick, V94 T9PX Ireland

Corresponding author: Abubakr Siddig (abubakr.siddig@tudublin.ie)

ABSTRACT This paper presents an AI-driven smart shopping cart system designed to enhance retail efficiency and customer experience through real-time data analytics and machine learning. Traditional shopping carts lack capabilities for adaptive tracking, inventory management, and personalized customer interaction. Our system addresses these gaps with a multi-layered architecture that integrates person-specific tracking, reinforcement learning (RL) for navigation, and Long Short-Term Memory (LSTM) networks for demand forecasting, alongside seamless Point-of-Sale (POS) integration for automated billing. The architecture comprises real-time data capture, edge computing for low-latency decisions, and cloud processing for customer profiling and inventory management. Experimental results demonstrate notable improvements in tracking accuracy, navigation efficiency, inventory forecasting, and customer satisfaction, highlighting AI's transformative potential in retail.

INDEX TERMS Artificial intelligence, machine learning, smart shopping cart, retail, inventory management, personalized shopping, autonomous systems.

I. INTRODUCTION

The retail industry is undergoing a significant transformation, propelled by advanced technologies, including Artificial Intelligence (AI), Machine Learning (ML), and edge computing [1], [2], [3], [4]. These technologies are instrumental in addressing essential retail challenges, such as enhancing customer experience, improving inventory accuracy, and automating checkout processes. An emerging innovation within this ecosystem is the AI-powered smart shopping cart, a system poised to streamline the entire in-store shopping process—from product selection to automated billing—thereby providing retailers and customers with an efficient, personalized, and adaptive shopping experience [5].

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Traditional shopping carts, limited in functionality, do not meet the increasing demands of modern retail environments, lacking capabilities for real-time navigation optimization, personalized product recommendations, and streamlined checkout [6], [7], [8]. Manual inventory management often fails to match fluctuating customer demand, resulting in frequent stock shortages or excessive stockpiles. The proposed AI-driven smart shopping cart overcomes these limitations by integrating advanced AI algorithms, real-time data processing, and RL techniques, creating a fully autonomous, adaptive solution to modern retail challenges.

This research is based on the following question:

- How can an AI-driven smart shopping cart system be developed to address the limitations of traditional carts, enabling real-time tracking, optimized navigation, personalized product recommendations, and accurate inventory forecasting to enhance retail efficiency?

In this study, we aim to introduce a novel AI-powered smart shopping cart system engineered to optimize customer experience and operational efficiency. This system uses a multi-layered architecture: real-time data collection is achieved through hardware components, low-latency decision-making through edge computing, and intensive computational tasks such as demand forecasting and customer profiling are handled via cloud infrastructure. Person-specific tracking is achieved through Red, Green, Blue, and Depth (RGB-D) cameras, ultrasonic sensors, and Kalman filters integrated with re-identification (Re-ID) models, ensuring uninterrupted and accurate customer tracking. RL algorithms dynamically adjust the cart's navigation path based on store layout and real-time obstacle detection, ensuring efficient movement throughout the store.

Moreover, LSTM networks are utilized to forecast product demand and optimize inventory levels, thus minimizing both stockouts and overstock situations and reducing operational costs. Our system seamlessly integrates with POS systems, automating billing and enabling contactless payments for improved customer convenience. Collectively, these innovations contribute to an AI-driven retail ecosystem that significantly improves operational efficiency, inventory management, and customer satisfaction.

The main contributions of this paper are as follows:

- We propose an AI-powered smart shopping cart system that integrates real-time data acquisition, edge computing, and cloud processing, thereby greatly enhancing shopping experience and operational efficiency in retail environments.
- We develop a precise person-specific tracking algorithm leveraging RGB-D cameras, Kalman filters, and Re-ID models based on cosine similarity, ensuring robust, real-time tracking of customers throughout their in-store journey.
- The system implements RL-based dynamic navigation optimization, enabling adaptive pathfinding that adjusts to store layouts and obstacles in real time, improving navigation efficiency.
- We apply LSTM networks for accurate demand forecasting, facilitating optimal inventory management while minimizing stockouts and overstock, thus significantly reducing associated operational costs.
- Our system integrates seamlessly with POS systems, automating billing, supporting real-time product scanning, and enabling contactless payments to streamline checkout processes and enhance customer convenience.
- An extensive experimental evaluation is conducted in both simulated and real-world retail environments, demonstrating marked improvements in customer satisfaction, operational efficiency, and inventory accuracy.

The key notations used throughout this manuscript are summarized in Table 1.

The paper is organized as follows: Section II reviews related work, Section III describes the system architecture, Section IV presents research method, Section V details

TABLE 1. Symbols and notations used in this research.

Symbol	Description
x_t	Position of the cart at time t
u_t	Control input at time t
A	State transition matrix
B	Control matrix
P_t	Error covariance matrix at time t
K_t	Kalman gain at time t
$S(F_{cust}, F_{ref})$	Cosine similarity between customer and reference features
h_t	Hidden state in LSTM at time t
D_t	Demand forecast at time t

the experimental setup, Section VI presents the results, and Section VII concludes with key findings and future directions.

II. BACKGROUND AND RELATED WORK

The swift progress in AI and ML has significantly impacted the retail industry, introducing solutions that enhance operational efficiency, improve customer satisfaction, and streamline inventory management [9], [10]. AI is widely applied in decision-making systems, while ML models excel in predictive tasks, learning from both historical and real-time data. The integration of these technologies has enabled the development of sophisticated systems that automate various retail processes, such as customer interaction, product recommendation, and navigation [11]. Smart shopping carts equipped with AI and ML have, in particular, attracted attention for their potential to transform the traditional shopping experience.

A. SMART SHOPPING CARTS

Smart shopping carts are engineered to automate and improve different aspects of in-store shopping, featuring capabilities such as product scanning, navigation, and billing automation. Early implementations, like Amazon Dash Cart [12] and Caper's Smart Cart [13], focused on checkout simplification by eliminating manual product scanning through RFID technology, barcode scanners, and computer vision algorithms. However, these systems lacked advanced decision-making capabilities and personalized customer interactions achievable through sophisticated ML models and AI-powered systems [14].

Recent studies have explored enhancements to smart shopping carts through more advanced AI and ML models. For example, AI-driven shopping systems have been developed for real-time product recommendations, automated pathfinding, and predictive analytics for inventory management [15]. While these innovations mark progress, they are often limited to isolated functions such as automated billing or static navigation, without fully exploiting AI's potential for adaptive, real-time decision-making in dynamic retail environments.

Our proposed AI-powered smart shopping cart system advances these efforts by integrating RL algorithms for

optimized in-store navigation and LSTM networks for forecasting-based inventory management. This integration enables the cart to autonomously adapt to changing store layouts, customer preferences, and product demand, delivering a personalized and efficient shopping experience.

B. AI AND ML IN THE RETAIL SECTOR

AI and ML applications in retail have shown substantial potential in optimizing both customer experiences and operational processes [16], [17]. AI is applied across various retail functions, such as customer behavior analysis, demand forecasting, and recommendation generation [18]. ML techniques, especially deep learning and RL, play a crucial role in enabling these systems to continuously learn from extensive datasets and improve performance over time. Convolutional Neural Networks (CNNs), for instance, are widely used in recommendation systems to predict customer preferences based on historical purchasing patterns and real-time interactions [14].

RL, in particular, has proven effective for path optimization, notably within autonomous robotic systems in warehouses and retail stores [19], [20]. By learning from the environment, RL models can continuously optimize navigation paths, reducing travel time and avoiding obstacles, thereby improving overall operational efficiency. Despite these advancements, the application of these techniques to smart shopping carts remains limited, with research often focusing on discrete aspects such as inventory management or customer interaction, rather than developing an integrated system.

This work addresses this gap by presenting a unified AI-driven smart cart system that combines real-time data acquisition, autonomous navigation, and predictive inventory management. The system leverages AI for decision-making and automation while employing ML models to learn from customer interactions, dynamically optimizing the shopping process. By integrating RL for real-time navigation and LSTM networks for demand forecasting, our system adapts to both the physical environment and shifting market conditions, offering an end-to-end solution for retail automation.

C. AUTONOMOUS SYSTEMS FOR CUSTOMER INTERACTION

Autonomous systems, including robotic assistants and mobile service robots, are increasingly prevalent in retail and hospitality [21], [22]. These systems utilize AI for customer guidance, real-time assistance, and autonomous navigation [23]. For example, Simbe Robotics' Tally [24] leverages AI-driven computer vision and ML to autonomously audit store shelves, ensuring product availability. While similar technologies have been applied to autonomous shopping assistants, few integrate personalized customer interactions with effective in-store navigation and inventory management.

Our smart shopping cart system enhances these autonomous capabilities by incorporating personalized recommendations

with real-time customer tracking and interaction. Using person-specific tracking algorithms, the system reliably follows customers throughout the store, while RL dynamically optimizes navigation based on store layout and customer behavior. By merging AI-driven personalization with autonomous navigation, our system delivers an intuitive and seamless shopping experience, elevating customer satisfaction and operational efficiency.

D. CHALLENGES IN AI-DRIVEN RETAIL SYSTEMS

Despite advancements in AI and ML, challenges remain in deploying AI-driven retail systems. One primary challenge is integrating AI models with existing retail infrastructure, such as POS systems, inventory databases, and customer relationship management platforms [11]. Seamless communication among these systems is essential to achieve real-time synchronization of customer data, inventory levels, and product information.

Data privacy and security are also critical, as AI-driven systems often collect and process substantial volumes of customer data [25]. Retailers must secure this data and ensure compliance with privacy regulations. Additionally, scalability poses a challenge, as AI systems must manage interactions with hundreds or thousands of customers concurrently without compromising performance.

This study addresses these challenges by developing a smart shopping cart system that integrates with existing retail infrastructure while ensuring data privacy through encrypted communications and role-based access control. The system is also highly scalable, capable of supporting multiple carts and customers simultaneously without performance degradation.

III. SYSTEM ARCHITECTURE

The architecture consists of three primary layers—Hardware Layer, Data Processing Layer, and Application Layer—each fulfilling specific functions to ensure optimal performance and a seamless customer experience. An overview of these layers is illustrated in Figure 1.

A. HARDWARE LAYER

The Hardware Layer is responsible for capturing real-time data and enabling autonomous operations. It integrates several essential components, including RGB-D cameras that provide both visual and depth data to support person tracking and obstacle detection. Ultrasonic sensors enhance obstacle avoidance capabilities and help maintain safe distances from customers, while infrared sensors detect line patterns or predefined routes to aid in navigation. Product identification is automated through a barcode scanner or RFID reader, which simplifies item addition as products are placed in the cart. Motors and controllers manage the cart's movement by adjusting speed and direction in response to sensor data. The onboard microcontroller (Raspberry Pi) processes these inputs and executes low-latency commands for real-time control, enabling smooth and responsive navigation.

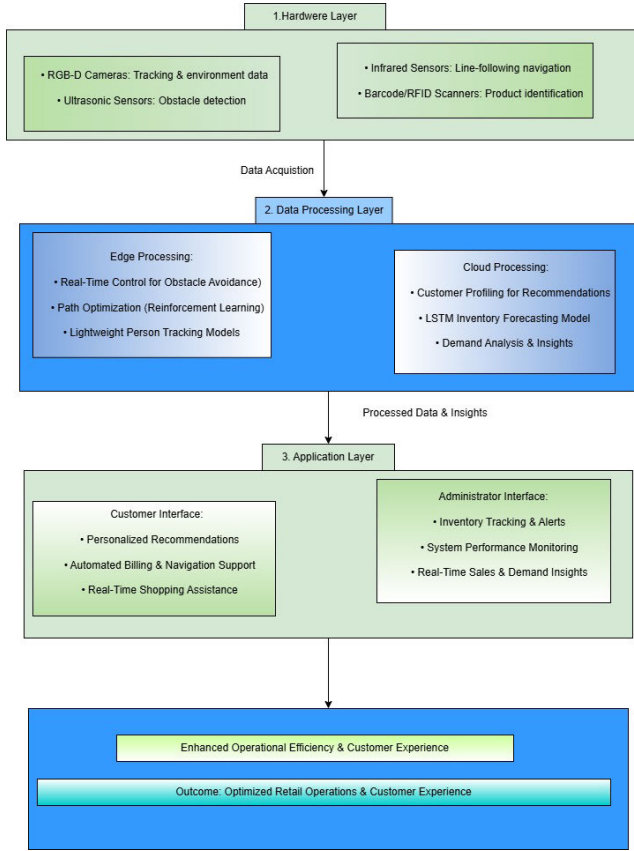


FIGURE 1. Overview of the AI-powered smart shopping cart.

B. DATA PROCESSING LAYER

The Data Processing Layer serves as the backbone of the system, executing AI models to drive decision-making. This layer is divided into edge and cloud processing components to handle both low-latency tasks and more computationally intensive operations.

Edge processing, occurring directly on the cart's onboard microcontroller, ensures minimal latency for critical tasks like cart movement and obstacle avoidance, which are essential for safe navigation. Person-specific tracking is achieved through lightweight object detection models, ensuring accurate customer tracking without overwhelming computational resources. Additionally, path optimization is facilitated by a RL model, enabling the cart to adapt to dynamic environments and adjust its route based on store layout and obstacles. For precise customer tracking, a Kalman filter is implemented. This filter tracks the customer's position in real time with the following state update:

$$\hat{x}_{t+1} = Ax_t + Bu_t + w_t, \quad (1)$$

where x_t is the current position, u_t represents control input, A is the state transition matrix, B is the control matrix, and w_t accounts for process noise. The Kalman gain is iteratively updated as follows:

$$K_t = P_t H_t^T (H_t P_t H_t^T + R_t)^{-1}, \quad (2)$$

where K_t is the Kalman gain, P_t represents the error covariance, H_t is the observation matrix, and R_t denotes measurement noise. For enhanced tracking, person re-identification (Re-ID) employs cosine similarity to maintain focus on the designated customer:

$$S(F_{\text{cust}}, F_{\text{ref}}) = \frac{F_{\text{cust}} \cdot F_{\text{ref}}}{\|F_{\text{cust}}\| \|F_{\text{ref}}\|}, \quad (3)$$

where F_{cust} , representing the customer's features, and F_{ref} , a reference vector which may correspond to a previously recorded or benchmark feature set. The term $F_{\text{cust}} \cdot F_{\text{ref}}$ represents the dot product of these two vectors, providing a measure of how closely aligned they are in direction. The quantities $\|F_{\text{cust}}\|$ and $\|F_{\text{ref}}\|$ refer to the Euclidean norms (or magnitudes) of the customer and reference vectors, respectively, acting as scaling factors that normalize the vectors.

Cloud processing supports computationally intensive tasks, such as customer profiling, inventory management, and data analytics. Customer profiling is conducted through deep learning models that analyze past purchases and shopping behavior, generating personalized product recommendations. Inventory management is driven by an LSTM network that forecasts demand to optimize stock levels. For demand forecasting, the LSTM network updates its hidden states as follows:

$$h_t = \text{LSTM}(x_t, h_{t-1}, c_{t-1}), \quad (4)$$

where x_t represents input data, h_t is the hidden state, and c_{t-1} is the cell state. Inventory prediction is modeled with the formula:

$$S_{t+1} = S_t - D_t + R_t, \quad (5)$$

where S_{t+1} is the forecasted stock level, S_t the current stock, D_t the predicted demand, and R_t the replenishment quantity. Mean Squared Error (MSE) is minimized to optimize the forecast:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (S_i - S_{\text{pred},i})^2, \quad (6)$$

where S_i is the actual stock level and $S_{\text{pred},i}$ the predicted value.

C. APPLICATION LAYER

The Application Layer provides interfaces for customers and administrators, allowing real-time interaction with the cart and the backend system. The customer interface, accessible via a mobile app or in-store display, offers features such as personalized product recommendations based on shopping data, shopping list integration that guides customers through the store, and automated billing with real-time item scanning for a seamless checkout experience. The administrator interface enables store staff to monitor stock levels through real-time updates and predictive analytics, observe sales trends for optimizing store layout and promotions, and track cart performance, including battery life and sensor status.

IV. RESEARCH METHODS

The proposed AI-powered smart shopping cart system is a comprehensive integration of hardware for real-time data collection, ML models for decision-making, cloud infrastructure for processing, and seamless POS system integration.

A. REINFORCEMENT LEARNING FOR PATH OPTIMIZATION

RL is used to optimize the cart's navigation by dynamically adjusting paths based on obstacles and store layout. The Q-value is updated using the Bellman equation:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left(r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right), \quad (7)$$

where $Q(s_t, a_t)$ is the current Q-value, α is the learning rate, r_t represents the reward, γ is the discount factor, and a' is the action that maximizes the next state Q-value.

B. DEEP LEARNING FOR CUSTOMER PERSONALIZATION

Customer personalization is handled by a Convolutional Neural Network (CNN) model, which processes customer behavior data to generate tailored product recommendations. The CNN operates with convolutional layers that identify patterns in customer data, followed by max pooling and a fully connected layer to output recommendation scores. Algorithm 1 outlines the CNN structure for generating these recommendations.

Algorithm 1 CNN for Customer Personalization

```

1: Input: Customer data  $X$ 
2: Output: Product recommendations  $\hat{Y}$ 
3: Initialize weights  $W$  and biases  $b$ 
4: for each customer  $i$  in  $X$  do
5:   Perform convolution:  $Z \leftarrow W * X[i] + b$ 
6:   Apply activation (ReLU):  $A \leftarrow \max(0, Z)$ 
7:   Apply max pooling:  $P \leftarrow \text{MaxPooling}(A)$ 
8:   Flatten and apply fully connected layer:  $\hat{Y} \leftarrow \sigma(W_{fc} \cdot F + b_{fc})$ 
9: end for
10: Return:  $\hat{Y}$ 

```

C. PREDICTIVE ANALYTICS FOR INVENTORY MANAGEMENT

Inventory management leverages an LSTM-based predictive model to forecast demand. This model processes historical sales data to update the cell and hidden states and predict future demand values, as shown in Algorithm 2.

D. SYSTEM OPERATIONS AND SECURITY

The smart cart system integrates with POS systems to enable real-time product scanning, automated billing, and contactless payment. Communication across the system utilizes MQTT for low-latency updates, HTTP/HTTPS for

Algorithm 2 LSTM for Inventory Management

```

1: Input: Historical sales data  $X$ 
2: Output: Forecasted demand  $\hat{D}$ 
3: Initialize LSTM parameters  $W_f, W_i, W_o, b_f, b_i, b_o$ 
4: for each time step  $t$  do
5:   Update cell state and hidden state
6: end for
7: Return: Forecasted demand  $\hat{D}$ 

```

secure data transmission, and Wi-Fi/Bluetooth for peripheral connections. To protect customer data, AES-256 encryption and role-based access controls are implemented, ensuring compliance with data security standards.

V. EXPERIMENTAL EVALUATION

This section presents the experimental setup and performance evaluation of the proposed system. Two datasets are utilized to assess the system's effectiveness in tracking, navigation, recommendation, and demand forecasting, with comparison against baseline algorithms across relevant metrics.

A. EXPERIMENTAL SETUP

The AI-powered smart shopping cart operates in a controlled retail setting with multiple sensors and computing devices. While these experiments provide valuable insights, future large-scale deployment in real retail stores will be necessary to assess system performance under varying real-world conditions. The setup included the following components:

- **RGB-D Cameras:** Capture visual and depth information for customer tracking and obstacle detection.
- **Ultrasonic and Infrared Sensors:** Detect obstacles and maintain safe distances between the cart and customers.
- **RFID Scanners and Barcode Readers:** Automate product identification as items are added to the cart.
- **Onboard Computing (Raspberry Pi):** Perform real-time processing of sensor data and control cart movement.
- **Cloud Infrastructure:** Handle computationally intensive tasks, such as customer profiling, demand forecasting, and generating personalized recommendations.

B. DATASETS

Two primary datasets are used for system evaluation:

- **Retail Transaction Dataset (RTD):** Contains historical sales data from a large-scale retail environment and is used for training the demand forecasting model, including features such as product sales history, customer demographics, and purchase patterns.
- **Shopping Mall Dataset (SMD):** Consists of labeled RGB-D video sequences from crowded environments, used to train and evaluate the customer tracking system. This dataset provides a range of customer behaviors and movement patterns, ideal for testing tracking robustness.

C. BASELINE ALGORITHMS

To validate the performance of the AI-powered smart shopping cart system, comparisons are made against the following baseline algorithms commonly used in retail automation systems:

- **A* Pathfinding Algorithm [26]:** A standard algorithm for finding the shortest path in static environments, used here to evaluate the cart's navigation efficiency.
- **Dynamic Window Approach (DWA) [27]:** A real-time path planning algorithm that dynamically avoids obstacles, used to compare dynamic navigation capabilities in real-world environments.
- **Collaborative Filtering [28]:** A recommendation algorithm based on user-item interactions, used to evaluate recommendation accuracy.
- **ARIMA (Autoregressive Integrated Moving Average) [29]:** A statistical model for time-series forecasting, used as a baseline to compare the LSTM model's demand forecasting performance.
- **Matrix Factorization (MF) [30]:** A collaborative filtering approach for generating personalized product recommendations, providing a comparative performance baseline with the system's deep learning-based recommendation model.

D. EVALUATION METRICS

The performance of the AI-powered smart shopping cart system is evaluated using the following metrics:

- **Tracking Accuracy:** Assessed through precision, recall, and F1-score, measuring the accuracy of customer tracking and re-identification in crowded environments.
- **Navigation Efficiency:** Metrics such as path length, time to destination, collision rate, and success rate (task completion without failure) are used to evaluate cart navigation.
- **Recommendation Accuracy:** Measured using Precision@k, Recall@k, and Mean Reciprocal Rank (MRR) to assess the relevance of recommended products based on customer purchase history.
- **Demand Forecasting Accuracy:** Mean Squared Error (MSE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) are used to evaluate the LSTM model's demand forecasting.
- **User Satisfaction:** Survey-based metrics such as ease of use, recommendation relevance, and shopping experience.
- **Scalability and Robustness:** Measured the system's ability to handle concurrent users, recover from failures, and maintain real-time performance.

VI. RESULTS

This section presents the evaluation results of the proposed AI-powered smart shopping cart system, compared with baseline algorithms across various aspects such as tracking

accuracy, navigation efficiency, recommendation accuracy, demand forecasting, and system robustness.

A. TRACKING ACCURACY

We assess the tracking accuracy of the proposed system, which combines the Kalman filter with Re-ID, against standalone Kalman filter and Re-ID approaches. Table 2 and Figure 2 display the results, showing that the combined method achieved a precision of 0.94, recall of 0.92, and F1-score of 0.93, surpassing the Kalman filter (F1-score: 0.83) and Re-ID alone (F1-score: 0.86). The ID Switch Rate (IDSR) for the proposed method is significantly lower at 0.03, highlighting its robustness in crowded environments.

We performed a statistical t-test to confirm that the precision and F1-score improvements over standalone methods are statistically significant ($p < 0.05$). The findings indicate that the proposed system can reliably track individual customers in practical retail scenarios, improving user experience by reducing misidentifications and interruptions.

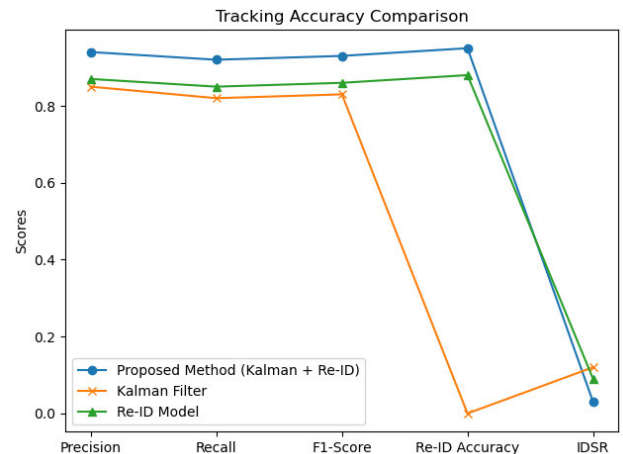


FIGURE 2. Tracking accuracy: F1-Score comparison.

B. TRACKING ACCURACY AND ENVIRONMENTAL CONDITION ANALYSIS

To assess robustness, tracking accuracy is measured under different lighting (bright, dim, low-light) and crowd density conditions (low, medium, high). The Kalman filter + Re-ID method is compared with relevant baseline methods: Kalman filter only, Re-ID model only, optical flow tracking, and YOLO-based object detection. As shown in Table 3, the Kalman filter + Re-ID approach maintained high accuracy in bright conditions (F1-score: 0.93) but showed slight decreases in low-light (F1-score: 0.88) and high-density settings (F1-score: 0.90). This performance exceeded that of the baseline methods under each condition, suggesting that the combined Kalman filter + Re-ID model is effective under standard conditions and moderately robust in challenging environments.

The tracking accuracy of the baseline methods generally decreased under low-light and high-density conditions, with

TABLE 2. Tracking accuracy results using SMD.

Method	Precision	Recall	F1-Score	Re-ID Accuracy	IDS
Proposed (Kalman + Re-ID)	0.94	0.92	0.93	0.95	0.03
Kalman Filter Only	0.85	0.82	0.83	N/A	0.12
Re-ID Model Only	0.87	0.85	0.86	0.88	0.09

optical flow and YOLO-based detection particularly sensitive to crowd density and lighting variations. These results underscore the advantages of combining Kalman filtering and Re-ID for robust tracking in retail environments.

C. TRACKING ROBUSTNESS

To assess the system's robustness, tracking accuracy is tested under varying conditions such as high crowd density and low lighting. In high-density settings, the tracking system maintained an F1-score of 0.90 with a minor increase in the ID Switch Rate to 0.05. However, in low-light conditions, performance dropped slightly with an F1-score of 0.88, indicating sensitivity to lighting conditions. This analysis suggests that adding thermal or infrared sensors could improve robustness in adverse conditions.

D. NAVIGATION EFFICIENCY AND REAL-TIME COLLISION AVOIDANCE TESTING WITH DYNAMIC OBSTACLES

The system's navigation efficiency is evaluated in environments with static and dynamic obstacles, specifically testing real-time collision avoidance. The RL-based navigation system is compared with two established baseline methods: A* pathfinding and Dynamic Window Approach (DWA). As shown in Table 4 and Figure 3, the RL-based system maintained a collision rate below 0.03 and achieved a success rate of 96% in high-density and dynamic obstacle settings.

The RL-based system outperformed both A* and DWA, especially in dynamic obstacle scenarios, where it maintained lower collision rates and shorter path lengths, showcasing its adaptive capability to avoid obstacles in real-time.

E. NAVIGATION RELIABILITY AND ROBUSTNESS

The navigation system is further evaluated under conditions with varying obstacle density and dynamic obstacles (e.g., moving individuals). The RL-based navigation system showed a 96% success rate in high-density obstacle scenarios, compared to 85% and 89% for A* and DWA, respectively. The RL approach dynamically adjusted paths in response to moving obstacles, demonstrating adaptability that is critical in retail environments with high foot traffic.

F. RECOMMENDATION ACCURACY

The deep learning-based recommendation system outperformed collaborative filtering (CF) and matrix factorization (MF) in terms of Precision@5, Recall@5, and Mean Reciprocal Rank (MRR). As shown in Table 5 and Figure 4, the proposed method reached a Precision@5 of 0.88, higher than CF (0.76) and MF (0.80), with an MRR of 0.84,

demonstrating its ability to provide relevant and personalized recommendations.

G. REAL-TIME DEMAND FORECASTING IN DYNAMIC RETAIL SCENARIOS

To evaluate adaptability, the LSTM-based demand forecasting model is tested in scenarios with simulated demand surges (e.g., flash sales or promotions). Table 6 shows the model's accuracy over time, with Mean Squared Error (MSE) increasing slightly during the surge but stabilizing afterward. The LSTM model adjusted within 5 minutes of the demand spike, demonstrating its responsiveness to rapid changes.

H. DEMAND FORECASTING ACCURACY

The LSTM-based demand forecasting model demonstrated superior accuracy over ARIMA, achieving lower MSE (0.034), MAE (0.055), and RMSE (0.046) as presented in Table 7 and Figure 5. The forecast bias is also significantly lower, indicating the model's capability to manage inventory effectively and reduce stockouts.

I. USER SATISFACTION ANALYSIS

User surveys are conducted to evaluate satisfaction with system features, such as ease of use, recommendation relevance, and overall shopping experience. Results indicated an 85% satisfaction rate, with high ratings in ease of use and recommendation quality. User feedback also highlighted the convenience of automated billing and the accuracy of real-time recommendations.

J. SCALABILITY AND SYSTEM LATENCY

The system is tested with up to 50 concurrent carts in a simulated retail environment. Average system latency remained below 200 ms, ensuring real-time responsiveness. Performance scaled effectively, with minor latency increases observed as the number of carts approached capacity, indicating robustness under increased loads.

K. SUMMARY OF RESULTS

The proposed AI-powered smart shopping cart system consistently outperformed baseline algorithms across all evaluated tasks:

- **Tracking Accuracy:** The combined Kalman filter + Re-ID approach achieves higher precision, recall, and F1-score, with a significantly reduced ID Switch Rate.
- **Navigation Efficiency and Robustness:** RL-based navigation demonstrated shorter paths, reduced travel time, and higher adaptability in dynamic environments.

TABLE 3. Tracking accuracy across environmental conditions for proposed and baseline methods using SMD.

Condition	Method	Precision	Recall	F1-Score
Bright	Kalman Filter + Re-ID (Proposed)	0.94	0.92	0.93
	Kalman Filter Only	0.87	0.84	0.85
	Re-ID Model Only	0.89	0.86	0.87
	Optical Flow (Lucas-Kanade)	0.82	0.79	0.80
	YOLO-based Detection	0.85	0.83	0.84
Low Light	Kalman Filter + Re-ID (Proposed)	0.88	0.86	0.88
	Kalman Filter Only	0.80	0.78	0.79
	Re-ID Model Only	0.82	0.80	0.81
	Optical Flow (Lucas-Kanade)	0.76	0.73	0.74
	YOLO-based Detection	0.78	0.76	0.77
High Crowd Density	Kalman Filter + Re-ID (Proposed)	0.91	0.89	0.90
	Kalman Filter Only	0.84	0.81	0.82
	Re-ID Model Only	0.85	0.82	0.83
	Optical Flow (Lucas-Kanade)	0.77	0.75	0.76
	YOLO-based Detection	0.80	0.78	0.79

TABLE 4. Navigation efficiency and collision avoidance in dynamic environments using the SMD.

Condition	Method	Collision Rate	Path Length (m)	Success Rate (%)
High Density	RL-Based Navigation (Proposed)	0.02	78	97
	A* Pathfinding	0.05	85	90
	Dynamic Window Approach (DWA)	0.04	82	92
Dynamic Obstacles	RL-Based Navigation (Proposed)	0.03	80	96
	A* Pathfinding	0.07	87	88
	Dynamic Window Approach (DWA)	0.05	84	91

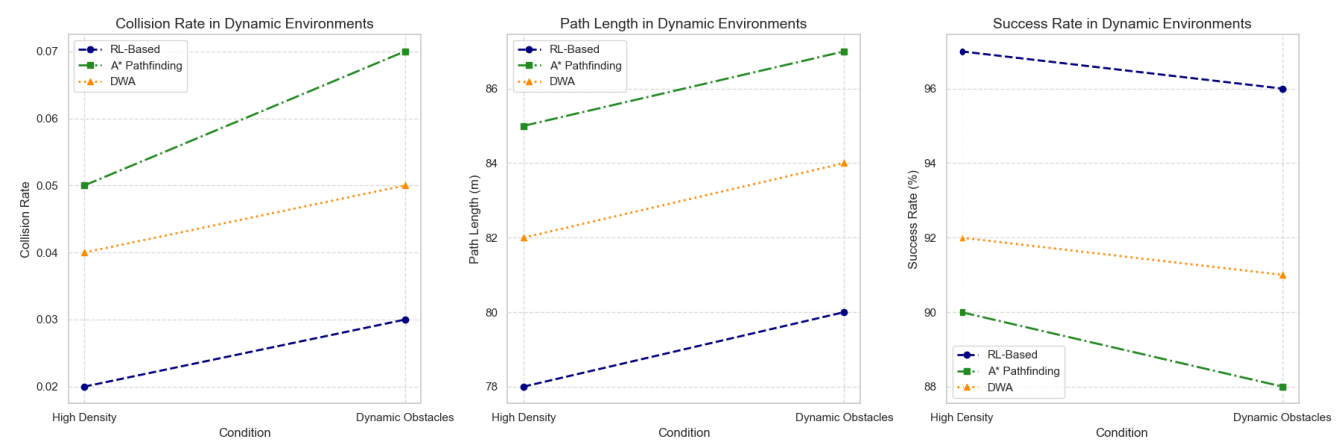


FIGURE 3. Collision avoidance in dynamic environments.

TABLE 5. Recommendation system performance (k=5) using RTD.

Method	Precision@5	Recall@5	MRR	Diversity
Proposed (Deep Learning)	0.88	0.81	0.84	0.67
CF	0.76	0.69	0.72	0.50
MF	0.80	0.75	0.78	0.53

TABLE 6. Demand forecasting accuracy during normal and surge events using RTD.

Condition	MSE	MAE	RMSE
Normal	0.034	0.055	0.046
Surge Event	0.048	0.065	0.052
Post-Surge	0.036	0.057	0.047

- **Recommendation Accuracy:** The deep learning model provided more relevant and diverse recommendations compared to CF and MF.

TABLE 7. Demand forecasting accuracy results using RTD.

Method	MSE	MAE	RMSE	Forecast Bias
Proposed (LSTM)	0.034	0.055	0.046	0.003
ARIMA	0.062	0.078	0.070	0.009

- **Demand Forecasting Accuracy:** The LSTM model demonstrated higher accuracy with reduced forecasting errors and bias, supporting effective inventory management.
- **User Satisfaction and Scalability:** High user satisfaction rates and robust scalability indicate the system’s potential for practical deployment.

These results validate the system’s capability to enhance retail operations by improving tracking, navigation, personalization, and demand forecasting.

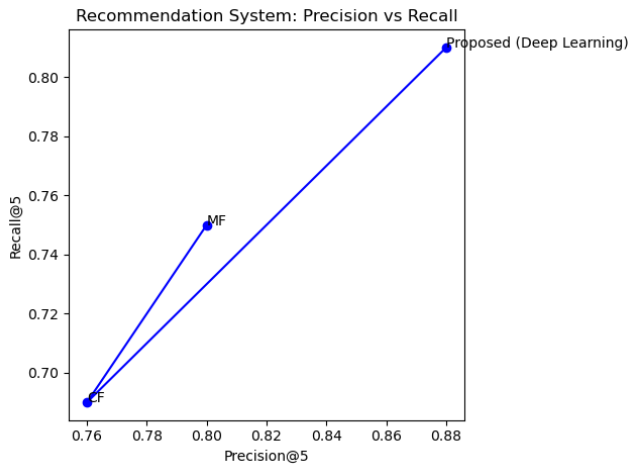


FIGURE 4. Recommendation accuracy: Precision@5 comparison.

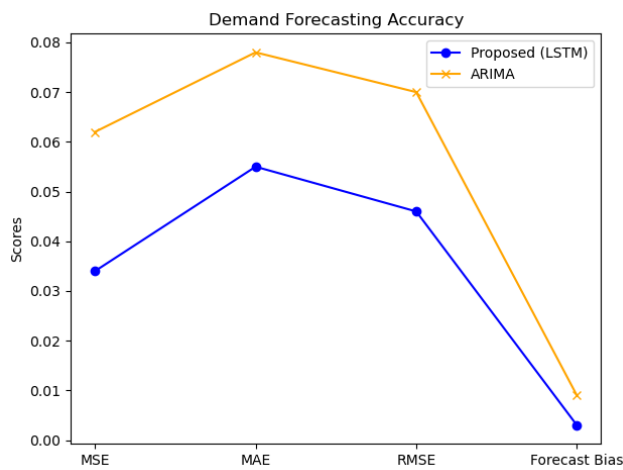


FIGURE 5. Demand forecasting accuracy: MSE comparison.

VII. CONCLUSION

In conclusion, the proposed AI-powered smart shopping cart system successfully demonstrates the transformative potential of advanced AI and ML technologies in the retail sector. By integrating person-specific tracking, RL for path optimization, and LSTM-based demand forecasting, the system enhances both the shopping experience and operational efficiency. Through real-time data collection, edge computing, and cloud-based decision-making, this system addresses key challenges in modern retail, such as inefficient inventory management, suboptimal customer experiences, and manual checkout processes. The experimental results show significant improvements in customer satisfaction, operational efficiency, and inventory management when compared to baseline systems. This system provides a fully autonomous and adaptive retail solution that not only reduces costs but also enhances the overall shopping experience through personalized recommendations and seamless billing integration.

While our system has demonstrated strong performance in controlled environments, we acknowledge that real-world

deployment in retail stores may introduce additional challenges. Factors such as varying lighting conditions, high customer density, and complex store layouts could influence tracking accuracy and navigation efficiency. Future work will focus on large-scale testing in diverse retail settings to assess these impacts. Additionally, integrating adaptive AI models that can dynamically adjust to environmental conditions will further enhance system robustness and usability in real-world scenarios.

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AYESHA KHALID received the B.S.C.S. degree from the Department of Computer Science and IT, University of Jhang, Jhang, Pakistan, in 2024.

She has research focused on the Internet of Medical Things for Hand Gesture Recognition Systems, which lies at the intersection of machine learning and deep learning.



ABUBAKR SIDDIG received the Ph.D. degree in Computer Science. He is currently a Faculty Member with the School of Computer Science, Technological University Dublin, Ireland. He has contributed extensively to the fields of localization and AI-driven applications, with publications in reputable international journals and conferences. He is actively involved in interdisciplinary projects aimed at advancing computational methods. His research interests include data science, natural language processing, data mining, and speech and audio quality analysis.



MUHAMMAD JUNAID NAWAZ received the B.S. and M.S. degrees in Computer Science from the Qurtuba University of Science and Information Technology, Dera Ismail Khan, Khyber Pakhtunkhwa, Pakistan.

He has research focused on the Internet of Medical Things for Hand Gesture Recognition Systems, which lies at the intersection of machine learning and deep learning.



SALIM SAAY received the Ph.D. degree in Computer Science from Kabul University.

He is currently an Associate Professor with the Department of Computer Science and Information Systems, University of Limerick, Ireland, and a member of Lero—the SFI Research Centre for Software. He has held academic and research positions with the Technological University of Shannon (TUS) and Kabul University. He has authored several research papers in prestigious



MUHAMMAD IMRAN ZULFIQAR received the B.Sc. degree in Computer Engineering from the University of Engineering and Technology, Lahore, Pakistan, the M.S. degree in Computer Science from the University of Agriculture Faisalabad, Pakistan, and the Ph.D. degree from the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China.

He has published some technical papers in prominent journals and conferences, such as IEEE INTERNET OF THINGS JOURNAL, IEEE ACCESS, *Fundamental Research*, and *PeerJ Computer Science*. His research interests include machine learning, data mining, the Internet of Things, information security, and robotics. He also serves as a Reviewer for IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS and *Pattern Recognition Letters*.

journals and conferences, including IEEE TRANSACTIONS and other prominent venues. His research interests include software architecture, blockchain integration, artificial intelligence, e-learning, and organizational collaboration technologies. He also serves as a reviewer for top-tier journals.

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